



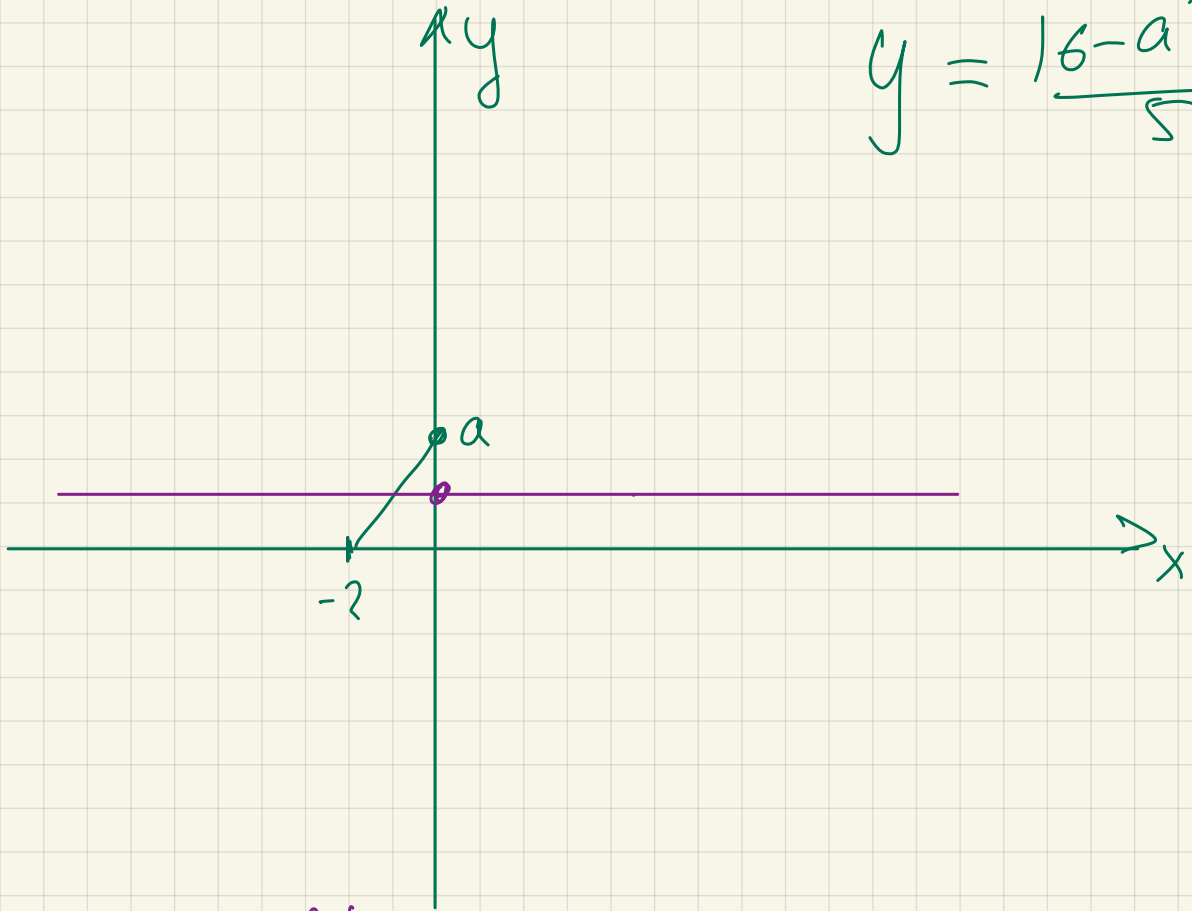
$$\begin{cases} \sqrt{(x+2)^2 + y^2} + \sqrt{x^2 + (y-a)^2} = \sqrt{4+a^2} \\ y = \frac{|6-a^2|}{5} \end{cases}$$

$$(-2; 0)$$

$$(0; a)$$

$\Rightarrow$

$$y = \frac{|6-a^2|}{5} \quad a \geq 0$$



$$0 \leq \frac{|6-a^2|}{5} \leq a$$

$$5 \cdot 0 \leq |6-a^2| \leq a \cdot 5$$

$\nearrow$

$|6-a^2| \geq 0$

$\nwarrow$

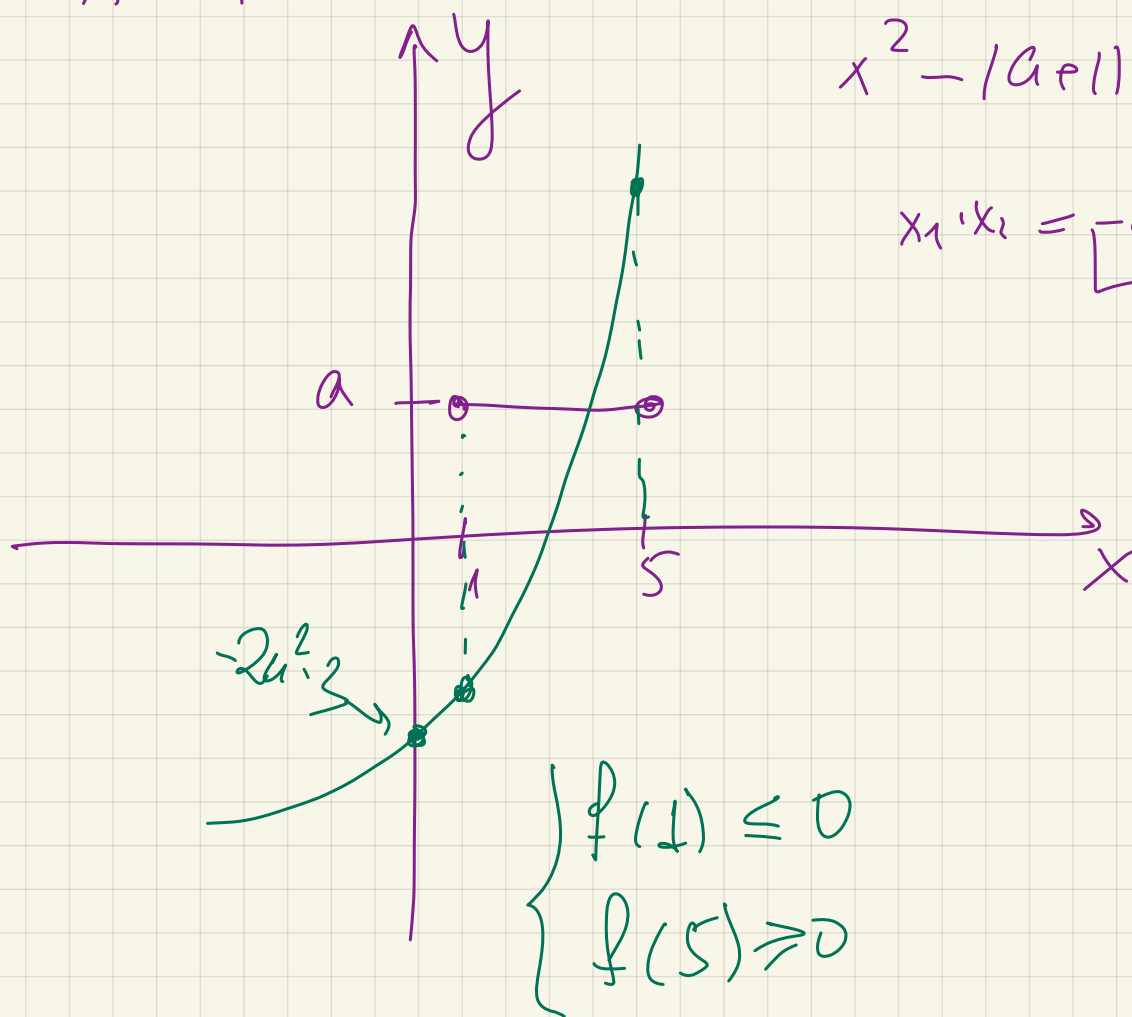
верно для любого a

$$|6-a^2| \leq 5a$$

$$\begin{cases} \sqrt{(x-1)^2 + (y-a)^2} + \sqrt{(x-5)^2 + (y-a)^2} = 4 \\ x^2 - |a+1| - 2a^2 = 3 \end{cases}$$

$$x^2 - |a+1| - \underbrace{2a^2 - 3}_{f(x)} = 0$$

$$x_1 \cdot x_2 = \underbrace{-2a^2 - 3}_{f(x)} < 0$$



$$\begin{cases} f(1) \leq 0 \\ f(5) \geq 0 \end{cases}$$

$$\begin{cases} 1 - |a+1| - 2a^2 - 3 \leq 0 \\ 25 - |a+1| - 2a^2 - 3 \geq 0 \end{cases}$$

$$2a^2 \leq -2$$

$$22 - 2a^2 \geq 0$$

$$\begin{aligned} & \cancel{\emptyset} \quad a^2 \leq 11 \\ & \quad \sqrt{11} \leq a \leq \sqrt{11} \\ & \quad a \geq -1 \end{aligned}$$

$$0 \leq -2a^2 - 2 \leq |a+1|$$

$$0 \leq 22 - 2a^2 \Rightarrow |a+1| \geq -2a^2 - 2$$

$$22 - 2a^2 \geq a+1 \geq -2a^2 - 2$$

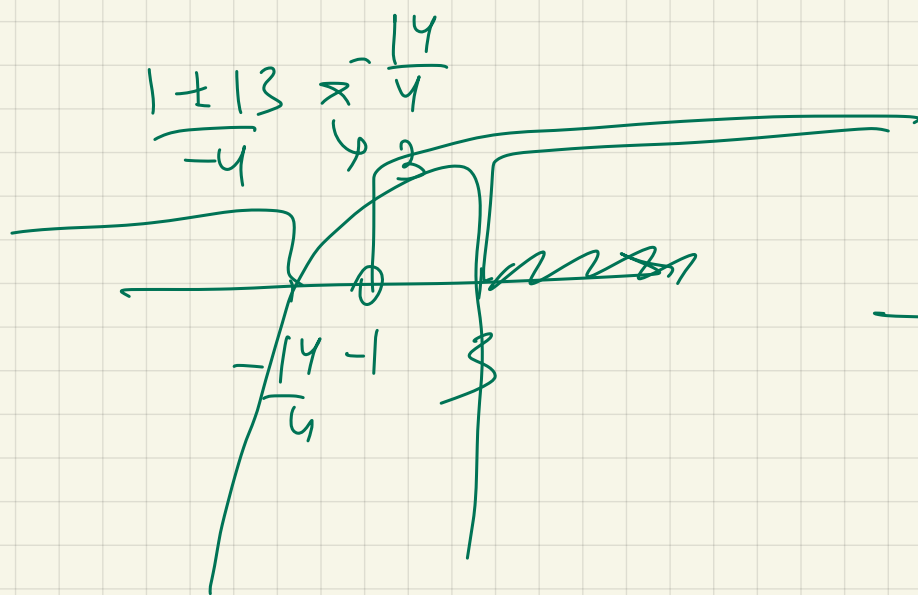
$$a \geq -2a^2 - 3$$

$$a \leq 22 - 2a^2 - 1$$

$$-2a^2 + 21 - a \geq 0$$

$$\frac{1 \pm \sqrt{1 + 2 \cdot 4 \cdot 2}}{-4}$$

$$\begin{array}{l} -2a^2 - 3 - a \leq 0 \quad \checkmark \\ 1 \pm \sqrt{1 -} \quad < 0 \\ \text{mod } a \end{array}$$



$$-a+1 \geq -2a^2-2$$

$$2a^2+1-a \geq 0$$

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$$-a-1 \leq -2a^2+22$$

$$2a^2-23+a \leq 0$$

$$\frac{-1 \pm \sqrt{1 + 23 \cdot 2 \cdot 4}}{4}$$

$$\frac{-1 \pm \sqrt{187}}{4}$$

$$\frac{-1 - \sqrt{187}}{4} \quad \frac{-1 + \sqrt{187}}{4}$$

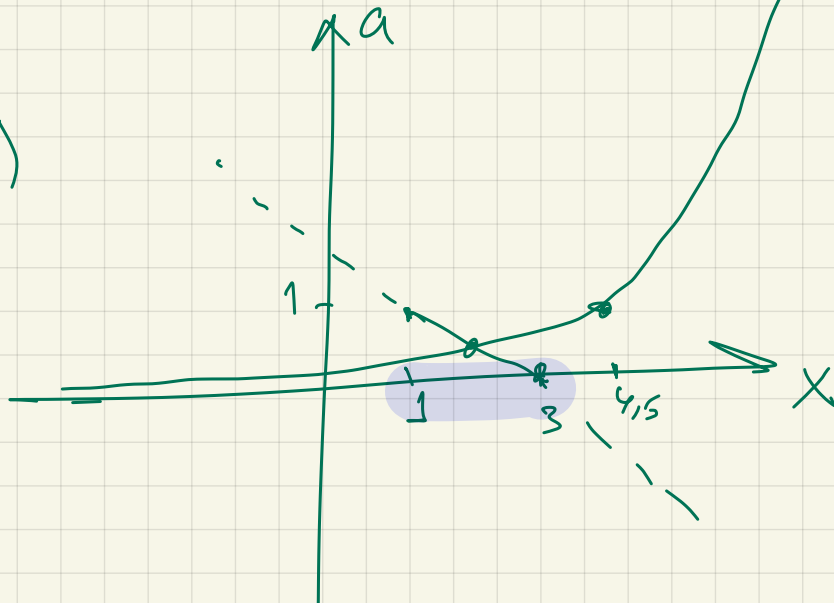
$$a \in \left[\frac{-1 - \sqrt{187}}{4}, -1 \right) \cup [3; +\infty)$$

$$a = 3$$

$$2^{1+x} = 32a\sqrt{2}$$

$$\sqrt{(x-1)^2 + (a-1)^2} + \sqrt{(x-3)^2 + a^2} = \sqrt{5}$$

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



$$a = \frac{2^{1+x}}{32\sqrt{2}} = 2^{1+x-5-1/2} = 2^{x-4,5}$$

$$\hat{a} = kx + b$$

$$1 = k + b$$

$$b = 1 - k$$

$$0 = 3k + b$$

$$\leftarrow 3k + 1 - k = 0$$

$$2k = -1$$

$$k = -\frac{1}{2}$$

$$b = 1 + \frac{1}{2} = \frac{3}{2}$$

$$a = -\frac{x}{2} + \frac{3}{2}$$

$$-\frac{x}{2} + \frac{3}{2} = 2^{x-4,5}$$

$$2^{-3} = \frac{1}{8}$$

$$\begin{matrix} x = 1,5 \\ x = 2,5 \end{matrix} \quad (+)$$

$$-\frac{3}{4} + \frac{3}{2} = \frac{3}{4}$$

$$-\frac{5}{2} \cdot \frac{1}{2} + \frac{3}{2} = -\frac{5}{4} + \frac{3}{2} = \frac{1}{4}$$

$$2^{2,5-4,5} = 2^{-2} = \frac{1}{4}$$

$$-\frac{6}{2} = -3$$

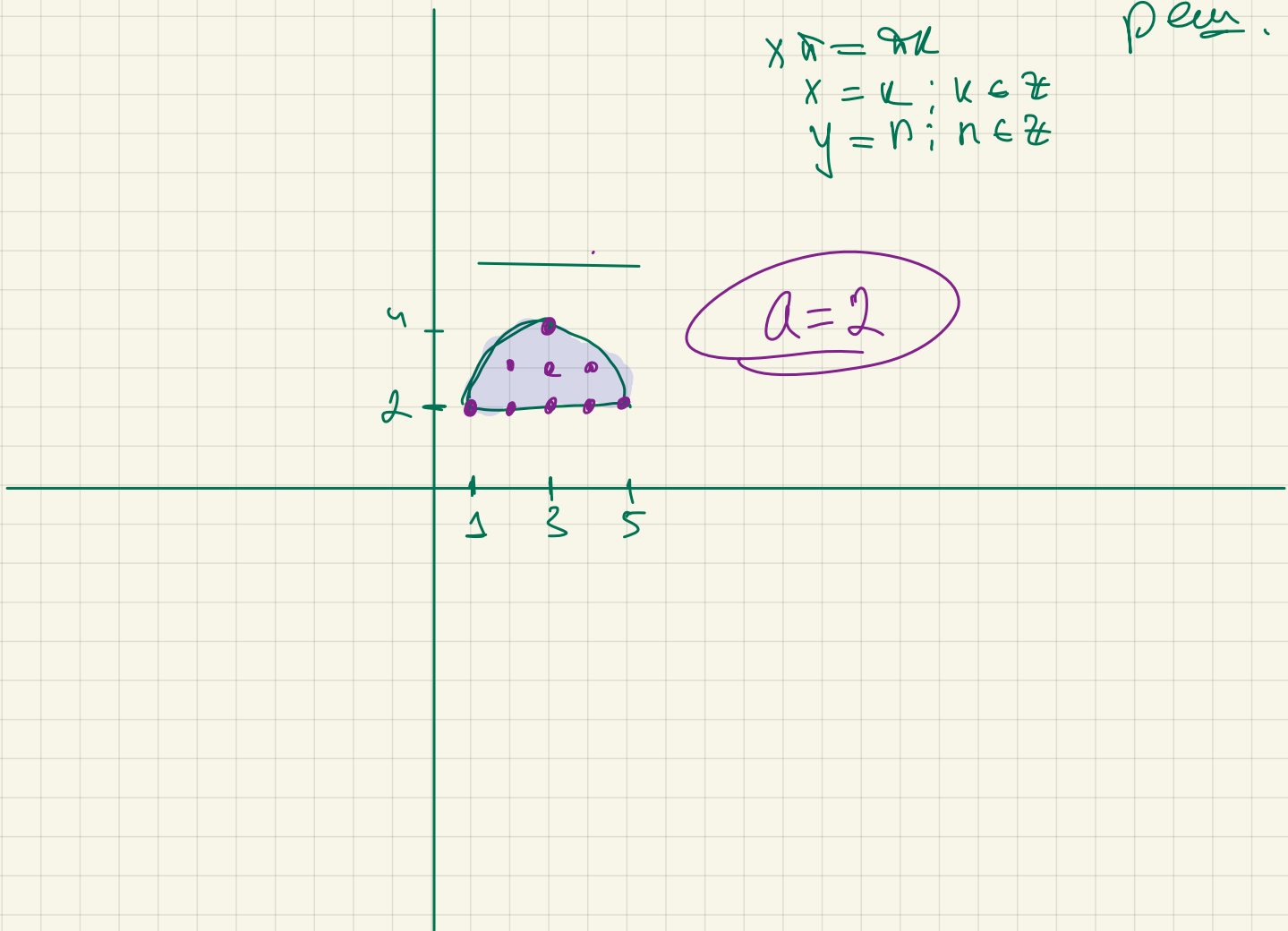
$$-9 + 18 - 5 = -x^2 + 6x - 5 \geq 0$$

$$\frac{-6 \pm \sqrt{36-20}}{-2}$$

$$\left\{ \begin{array}{l} 2 \leq y \leq 2 + \sqrt{6-x^2-5} \\ \sqrt{(x-1)^2 + (y-a)^2} + \sqrt{(x-5)^2 + (y-a)^2} = 4 \\ \sum m(x_i) = 0 \leftarrow \\ \sum m(y_i) = 0 \leftarrow \end{array} \right.$$

Haus. Kon.-bo
per.

$$\begin{aligned} x &= k \\ x &= k; k \in \mathbb{Z} \\ y &= n; n \in \mathbb{Z} \end{aligned}$$



a ; наиб. ≥ 1 и меньше $\rightarrow$
 $f(x) = |x-a| - x^2$

1) $f(x) = \underline{x-a-x^2}$

$x \geq a$

2) $f(x) = a-x-x^2$ $x < a$

1) $f'(x) = 1-2x = 0$ $x = \frac{1}{2}$

$f(\frac{1}{2}) = \frac{1}{2} - a - \frac{1}{4} \geq 1$

$a \leq \frac{1}{4} - 1 = -\frac{3}{4}$

$a \in (-\infty; -\frac{3}{4}]$

2) $-1-2x=0$ $x = -\frac{1}{2}$

$a + \frac{1}{2} - \frac{1}{4} \geq 1$

$a \geq 1 - \frac{1}{4} = \frac{3}{4}$

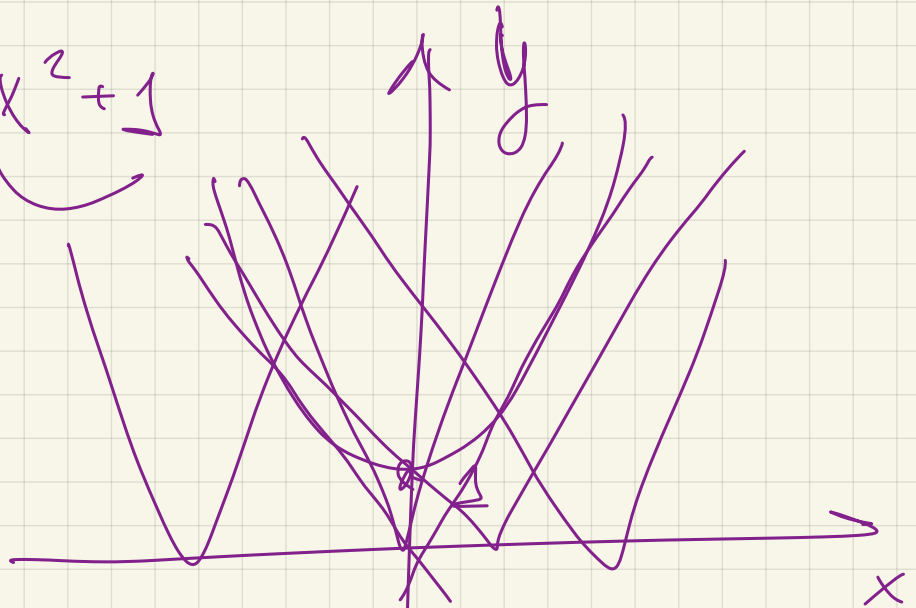
$$a \in (-\infty; -\frac{3}{4}] \cup [\frac{3}{4}; +\infty)$$

$$|x-a| - x^2 \geq 1$$

$$|x-a| \geq x^2 + 1$$

$f(x)$

$$y = |x-a|$$



a ; Наму з ш. Φ -м. $\geq \underline{\underline{-24}}$

$$f(x) = 4ax + |x^2 - 6x + 5|$$

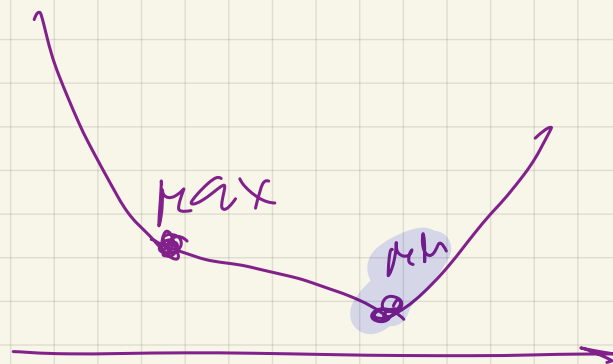
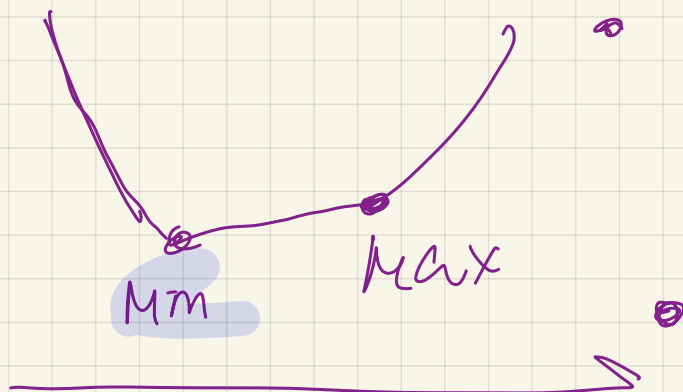
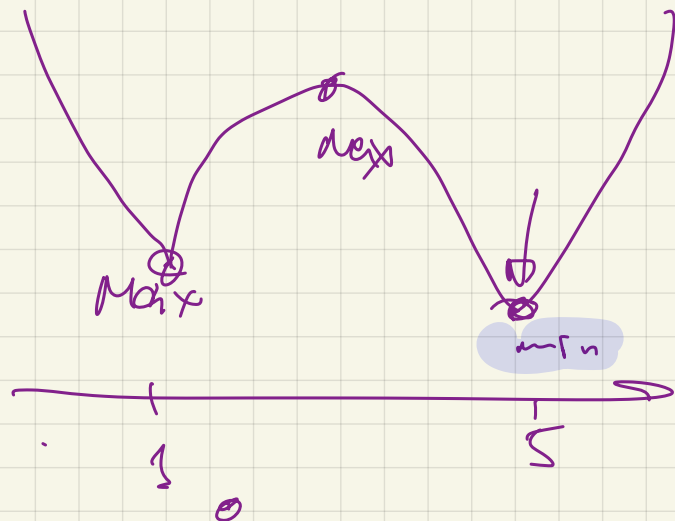
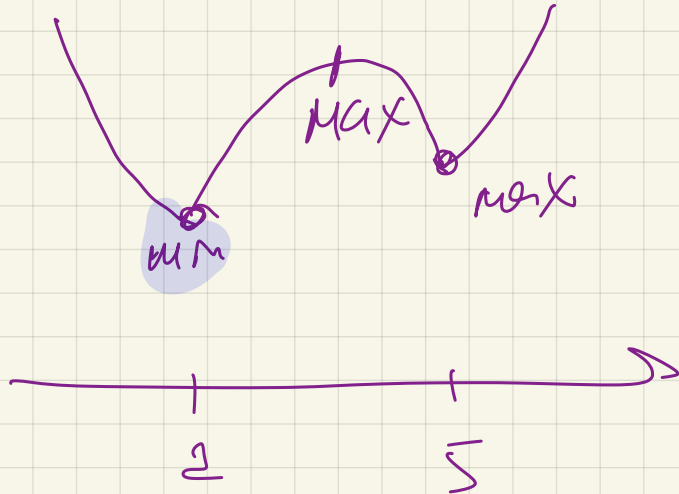
$$f(x) = 4ax + x^2 - 6x + 5$$

$$x \in (-a; 1] \cup [5; +\infty)$$

$$f'(x) = 4a + 2x - 6 = 0$$

$$x_0 = \frac{6-4a}{2} = 3-2a$$

$$3-2a \in [1; 5]$$



$$3-2a \in (-\infty; 1] \cup [5; +\infty)$$

$$\begin{cases} 3-2a \in [1; 5] \\ f(1) > -24 \\ f(5) > -24 \end{cases}$$

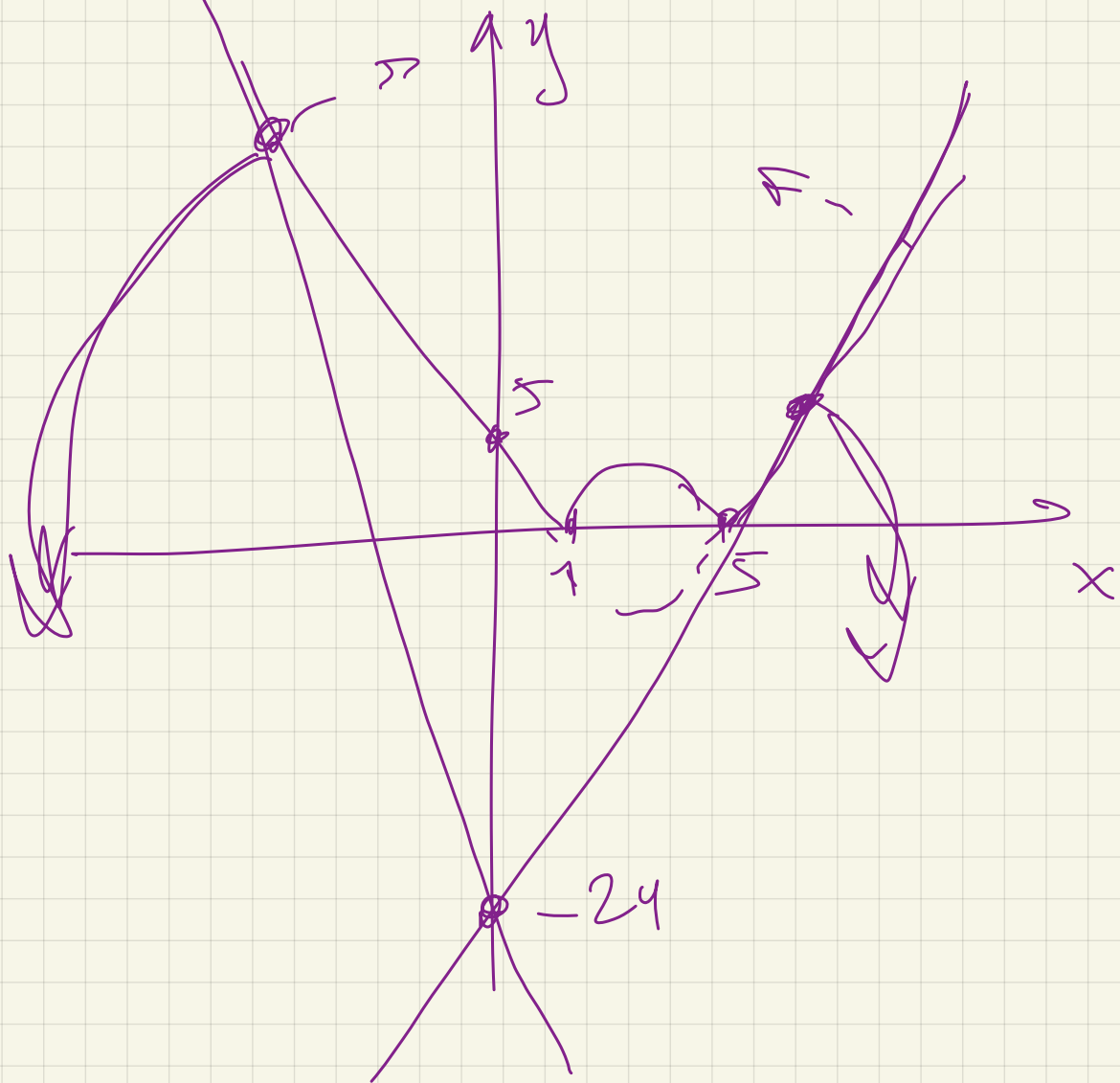
$$\begin{cases} 3-2a \in (-\infty; 1] \cup [5; +\infty) \end{cases}$$

$$\begin{cases} \underline{f(3-2a)} > -24 \end{cases}$$

$$f(x) = 4ax + |x^2 - 6x + 5| > -24$$

$$y = |x^2 - 6x + 5|$$

$$y = -24 - 4ax <$$



$$x^2 - 6x + 5 = -24 - 4ax$$

$$x^2 - 6x + 4ax + \underline{\underline{29}} = 0$$

$$\Delta = (-6+4a)^2 - 4 \cdot 24 = 0$$

$$(4a-6)^2 = 4 \cdot 24$$

$$4a-6 = \pm 4\sqrt{6}$$

$$4a = 6 \pm 4\sqrt{6}$$

$$a = \frac{6 \pm 4\sqrt{6}}{4}$$

$$a = \frac{3 \pm 2\sqrt{6}}{2}$$

$$a \in \left(\frac{3-2\sqrt{6}}{2} ; \frac{3+2\sqrt{6}}{2} \right)$$